PROFESSIONAL EDUCATION



AI AND ML: LEADING BUSINESS GROWTH



THE MIT PROFESSIONAL EDUCATION ADVANTAGE

The mission of MIT Professional Education is to provide a gateway to renowned MIT research, knowledge, and expertise for those engaged in science and technology worldwide, through advanced education programs designed for working professionals.

MIT Professional Education supports MIT's vision, including connecting practitioner-oriented education with industry. MIT Professional Education provides continuing education courses and lifelong learning opportunities for science, engineering, and technology professionals at all levels, from around the world. Worthy of note is that MIT professors and lecturers lead and teach all MIT Professional Education offerings. MIT Professional Education fosters the development of innovative leaders equipped to address complex problems globally.

PROGRAM OVERVIEW

96% of industry professionals say that they plan to use AI simulations, such as digital twins.
(Source: PwC 2022 AI Business Survey)

98% of companies think AI can help overcome labor market challenges.
(Source: PwC 2022 AI Business Survey)

82% of leaders said that their employees believe that working with AI technologies will enhance their performance and job satisfaction.
(Source: State of AI in the Enterprise Fifth Edition 2022)

Leading global organizations need business and technology leaders who can confidently navigate the fast evolving landscape in Artificial Intelligence (AI) and Machine Learning (ML) with a strategic perspective and ensure success through business applications. Knowledge and skills in application of AI and ML can help drive business growth, secure competitive advantage and ensure continued career progression.

The "AI and ML: Leading Business Growth" program by MIT Professional Education will arm you with the knowledge, tools, and best practices needed to lead strategic initiatives aimed at leveraging cutting-edge AI and ML to drive innovation, efficiency and business growth.

While you explore the critical aspects of AI and ML, the program's focus is on empowering you, as a business leader, to leverage the knowledge to work on efficiency, solution selection, implementation, risk management, and the impact on driving business growth.

You will gain a much better understanding of how to leverage AI and ML to benefit the future of your business by effective planning, and alignment with the strategic goals of the organization. You also will come to understand challenges inherent in deploying these exponential technologies and learn how to address them.

This AI and ML program is a comprehensive 6-month action-learning-based live virtual program led by MIT faculty. You will benefit directly from their expertise and research-driven insights as you learn, experiment, implement, and reflect, while comfortably managing your work responsibilities from any geographical location around the world.



PROGRAM FEES
USD 9,000



PROGRAM LOCATION
LIVE VIRTUAL

PROGRAM BENEFITS

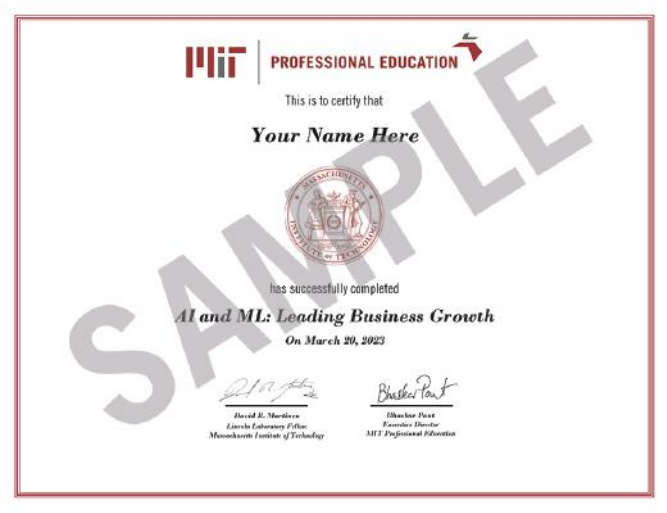
After the completion of this program, you will be a part of a cohort of global leaders and practitioners in the field of AI and ML.

You will also receive the benefits listed below:

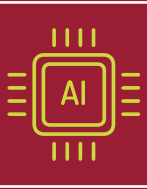
- You will become a member of the MIT Professional Education alumni network and join the exclusive MIT PE LinkedIn group
- Become a part of a cohort of global leaders and practitioners in the field of AI and ML
- Access MIT Professional Education sponsored events, webinars, and networking opportunities
- Enjoy One-year digital subscription to MIT Horizon
- Access updates on faculty research, new programs, and MIT initiatives via our communications
- Enjoy fifteen percent (15%) discount on enrollment in future MIT Professional Education offerings (Short duration open enrollment programs of 5 days or less and online courses)

*Benefits are subject to change and some may require an additional fee.

Upon the completion of the program, you will receive the AI and ML: Leading Business Growth certificate from MIT Professional Education.



KEY TAKEAWAYS



Gain a sound understanding of the strategic applications of AI and ML in business

Gain the tools needed to develop a strategic vision and development plan focused on AI products and services





Understand end-to-end AI architecture at the system engineering level

Master the AI pipeline building blocks for developing AI system capabilities





Discover the opportunities, capabilities, and scope of this transformative technology

Develop frameworks to analyze and monitor the use of AI and ML in your own organization





Communicate value proposition of AI and ML to stakeholders effectively to ensure success

Benefit from the “voice of practitioners” across various industries





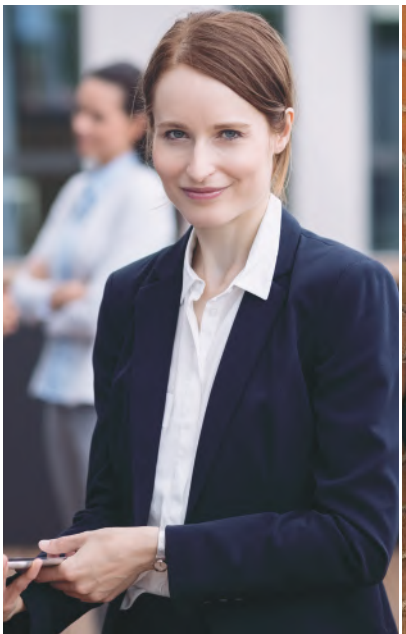
Understand the challenges and risks associated with AI and ML implementations

WHO SHOULD ATTEND

The AI and ML program is designed for high-performing business leaders, mid to senior level managers, consultants, and other business professionals. These dynamic individuals look to lead their organizations' success through strategic implementation and usage of AI and ML, driven by a deep understanding of technology, best practices, frameworks, and challenges.

The program is designed for working professionals with:

- **A minimum of 5 years of work experience, a demonstrated career progression, and success across levels, in roles such as:**
  - **Business leaders** such as Chief Digital Officers, Chief Technology Officers, Chief Strategy Officers responsible for leading digital transformation projects
  - **Domain leaders** such as directors of marketing, technology, innovation, or strategy responsible for creating business value by leveraging the power of AI and ML
  - **Technical leaders, product directors,** infrastructure heads, and solutions architects responsible for leading enterprise IT, infrastructure and product roadmaps
  - **Product managers, project managers, and mid-level managers** who strive to enhance efficiencies and user experiences with AI and ML driven automation, engagement, and insights
- Experience in leading teams and managing projects/ products/ team, preferred but not required
- An undergraduate degree with a strong academic track record is preferred
- Fluency in written and spoken English



PROGRAM EXPERIENCE

Through this AI and ML program, MIT Professional Education provides a tailored world-class educational experience for professionals from around the globe looking to advance their careers and creatively address complex problems.

MIT faculty will guide you to understand the current and future capabilities of this transformative technology, in order to effectively unlock its potential within business. You will also have the opportunity to design a roadmap for a real-world business case – tailored for your own industry. At the end of the program, you will be better prepared for immediate and practical business action.

The program does not require coding in python or R or any such language. However, it requires the ability to be agile with data. Participants are strongly encouraged to think about the strategic use cases that they would like to address. Above all, a positive attitude and curiosity to learn is of utmost importance.



LIVE VIRTUAL LECTURES

Interactive sessions with renowned MIT faculty.



PRACTITIONER INSIGHTS

Leading AI and ML practitioners will share first-hand experiences and lessons learned in the application of AI and ML in business.



IN-CLASS EXERCISES

Highlight key concepts discussed in the sessions.



TEAM PROJECTS

Focused on the strategic roadmap for an AI product or service for application-oriented learning.



NO CODING REQUIRED

Hands-on learning experience with a “no code” approach.



OUTSTANDING PEER GROUP

Diverse group of like-minded professionals that will elevate the learning environment and create lasting professional relationships.



DESIGNED FOR BUSY EXECUTIVES

Managed learning journey of 4-6 hours/week.



POLLS AND READING ASSIGNMENTS

Live sessions, assigned knowledge quizzes, and class projects enable students to gain broad and in-depth knowledge.

Here are some of the tools that you will learn to use in this program.

RAPIDMINER

ikigai

Teachable Machine

DISTINGUISHED FACULTY

All MIT Professional Education instructors are MIT professors and senior lecturers. You will learn from and network with prominent scholars, many of whom have founded their own companies and make headlines across the globe with their accomplishments.

The program is led by:



DAVID MARTINEZ

MIT Lead Instructor and Laboratory Fellow Cyber Security and Information Sciences Division, MIT Lincoln Laboratory

David R. Martinez is a Laboratory Fellow in the Cyber Security and Information Sciences Division at MIT Lincoln Laboratory. In this capacity, he is focusing on research and technical directions in the areas of artificial intelligence (AI), high-performance computing, and digital-enterprise transformation. Previously, Mr. Martinez served as an Associate Head in the Cyber Security and Information Sciences Division. He was also a member of Lincoln Laboratory's Steering Committee.

Mr. Martinez has held many past Laboratory leadership roles, including Leader of the Embedded Digital Systems Group and Head of the ISR Systems and Technology Division. Mr. Martinez also served in a leadership role as President and Chairman of Mercury Federal Systems.



DEVAVRAT SHAH

Professor of Electrical Engineering and Computer Science, MIT Director, Statistics and Data Science Center

Devavrat Shah is a professor with the department of electrical engineering and computer science, MIT. He is a member of the Laboratory for Information and Decision Systems (LIDS) and Operations Research Center (ORC), and the Director of the Statistics and Data Science Center (SDSC) in IDSS. His research focus is on theory of large complex networks, which includes network algorithms, stochastic networks, network information theory and large-scale statistical inference. Prof. Shah was awarded the first ACM SIGMETRICS Rising Star Award 2008 for his work on network scheduling algorithms. He received the 2010 Erlang Prize from INFORMS, which is given to a young researcher for outstanding contributions to applied probability. He is currently an associate editor of Operations Research.

Prof. Shah is also the lead instructor for multiple programs such as the Applied Data Science Program and Machine Learning: From Data to Decisions. He co-founded Celect, Inc, now part of Nike in 2013, to help retailers redesign store layouts by accurately predicting demand using omnichannel data. In 2019, he co-founded IkigaiLabs (ikigailabs.io) - a no-code AI platform where he serves as CTO, with the mission to build self-driving organizations by enabling data-driven operations with human-in-the-loop.

CURRICULUM

The program curriculum is uniquely designed to transform high-achieving professionals into strategic leaders with the skills and insights to transform their businesses and industries with the power of AI and ML. The program curriculum focuses on the application of key concepts and frameworks in the context of industry and business to give you a definitive competitive advantage.

You will gain a comprehensive understanding of the systemic aspects and practical application of AI and ML as you dive deeper into six main building blocks to develop transformational AI and ML capabilities.

1. CREATING A STRATEGIC VISION AND ROADMAP
2. TRANSFORMING DATA INTO INFORMATION
3. IDENTIFYING THE RIGHT COMPUTING AND INFRASTRUCTURE FRAMEWORKS FOR THE CONTEXT
4. APPLYING ALGORITHMS – TURNING INFORMATION TO KNOWLEDGE
5. HUMAN-MACHINE TEAMING – TURNING THE KNOWLEDGE INTO INSIGHTS
6. SUCCESSFUL DEPLOYMENT – ACHIEVING ROBUSTNESS AND TRUSTWORTHINESS, AND ADDRESSING RISKS



CURRICULUM

KEY MODULES

AI BACKGROUND AND SYSTEM ARCHITECTURE OVERVIEW

You will be taken through a brief history of AI, while diving deep into the AI domains of impact. You will also be introduced to the AI system architecture and address the key building blocks that form an end-to-end AI pipeline. You will master the challenges and key building blocks necessary for developing AI system capabilities.

AI STRATEGIC VISION AND IMPLEMENTATION ROADMAP

You will be introduced to a strategic development model. Using this framework, you will learn to address the formulation of a strategic vision, product and/or service value proposition, and the important components of an AI development roadmap. Overall, you will learn to formulate a strategic development plan that serves as the guiding blueprint for AI designers, developers, and customers.

UNDERSTANDING DATA, CONDITIONING, AND PREPARING DATA

In building AI and ML capabilities, a significant amount of time is spent conditioning the data to be ready for the application of ML algorithms. You will learn to architect systems to transform structured and unstructured data into information. You will gain the tools and know-how required to recognize the vulnerability of a system to data bias.

INTRODUCTION TO MACHINE LEARNING AND STRATEGIC APPLICATIONS

You will be introduced to machine learning and gain a deeper understanding of different methods of machine learning. You will learn how the different classes of ML algorithms transform information into knowledge. Here you will gain experience developing a simple neural network as well.

IMPLEMENTING MACHINE LEARNING IN A BUSINESS – PLANNING, MONITORING, EVALUATING AND CONSTRAINTS

Recognize and learn how you need to pivot everything from work processes to thought processes and integrate machine learning into the planning, monitoring, evaluating and implementation of business at every level. You will also learn to gauge the limitations and challenges associated with machine learning in business and how you can address and resolve those issues.

CURRICULUM

PREDICTIVE ANALYSIS AND DECISION SYSTEMS

Build the dexterity needed to extract insights from data and subsequently to make decisions that impact day-to-day mission-critical operations, as well as for long-term. Develop a deep understanding and hands-on experience of different methods to build decision systems and predictive capabilities.

MODERN COMPUTING FRAMEWORKS

Understand how recently developed computing technologies or domain specific architectures (DSAs) such as the Graphics Processing Units (GPUs) and the Tensor Processing Units (TPUs), enable narrow AI to be more effective. Understand what makes GPUs and TPUs well-matched with executing machine learning algorithms. Develop the capability to assess various computing technologies suited to your needs in terms of computation throughputs.

HUMAN-MACHINE TEAMING

Learn how human-machine teaming transforms the output of the ML algorithms from knowledge into insight for users to leverage the output of the AI and ML systems. You will develop the skills needed to classify areas where the application of AI augments the capabilities of humans working together as teams with the machine (or chiefly computer).

BUILDING AND DEPLOYING ROBUST AI CAPABILITIES

Develop the ability to assess the robustness of the AI system and ascertain the trustworthiness throughout its life cycle to be able to successfully deploy an AI system. You will learn about assessment of the sensitivity of ML algorithms to modifications of data and the algorithm model itself. You will also be introduced to a set of ten guidelines that involve people, processes, and technologies, and are necessary in the design, development, and implementation of end-to-end AI capabilities.

COMMUNICATING VALUE AND MANAGING STAKEHOLDERS

As C-suite executives, product managers, implementation and business leaders, it is important to be able to communicate the value proposition of AI products or services. You will master the skills and techniques, based on a proven methodology, to communicate the value proposition of AI products or services clearly and concisely.

CURRICULUM

CHALLENGES AND FUTURE OF AI AND ML

You will confront the possible challenges that you may encounter through your journey of transforming your business with AI and ML. You will also explore the predicted future potential of these exponential technologies, in the context of business growth.

ACTION LEARNING TEAM PROJECTS

The program embodies the MIT motto of "Mens et Manus" – Mind and Hand – the belief that academic knowledge must go hand in hand with practical application, and a very strong commitment to applied learning. You will work on team assignments and projects to apply the learning to a strategic business problem on an on-going basis throughout the program. It is a unique opportunity for collaborative learning and applying the knowledge of technologies, frameworks, and best practices to real-world problems.

ON-DEMAND OPEN HOUSE

AI and ML: Leading Business Growth Program
from MIT Professional Education



DAVID MARTINEZ

MIT Lead Instructor and Laboratory Fellow Cyber Security and Information Sciences Division, MIT Lincoln Laboratory



DEVAVRAT SHAH

Professor of Electrical Engineering and Computer Science, MIT Director, Statistics and Data Science Center

WATCH NOW

Scan for on-demand open house session for MIT Professional Education AI and ML: Leading Business Growth program

ADMISSIONS

The “AI and ML: Leading Business Growth” program follows a rolling process (earlier application, earlier consideration) and participant applications are evaluated as soon as they are received. The step-by-step process is outlined below.

ONLINE APPLICATION

You can apply to the program online, by submitting a short application form. We do not accept any paper applications.

RESUME

Keep a resume in **Word or PDF format** ready to be submitted (via upload) through the upload tab in the online application form.

REFERENCES

We require **one professional reference** who is ready to discuss your candidacy with us in great detail.

Click the link below to apply to the program:
https://northwest.education/mitpe-aiml/apply

PROGRAM FEE

The program fee is USD 9,000 inclusive of all taxes. It covers the teaching fee, usage of the learning platform, and all online learning materials. It does not cover any other expenses that are not expressly mentioned above.

Considering the limited seats, we encourage you to apply as early as possible. Early registration benefits might also be available to applicants who apply in early application rounds. Please click here to explore the benefits available or connect with a program advisor.



ABOUT MIT

Founded in 1861, MIT is committed to generating, disseminating, and preserving knowledge, and to working with others to bring this knowledge to bear on the world's great challenges. MIT is dedicated to providing an education that combines rigorous academic study and the excitement of discovery with the support and intellectual stimulation of a diverse campus. MIT is profoundly global and the community gains tremendous strength as a magnet for talent from around the world. Through teaching, research, and innovation, MIT's exceptional community pursues its mission of service to the nation and the world. MIT seeks to develop in each member of the MIT community the ability and passion to work wisely, creatively, and effectively for the betterment of humankind.

MIT is an integral part of its host city of Cambridge, a diverse and vibrant community noted for its intellectual life, history, and thriving innovation climate.

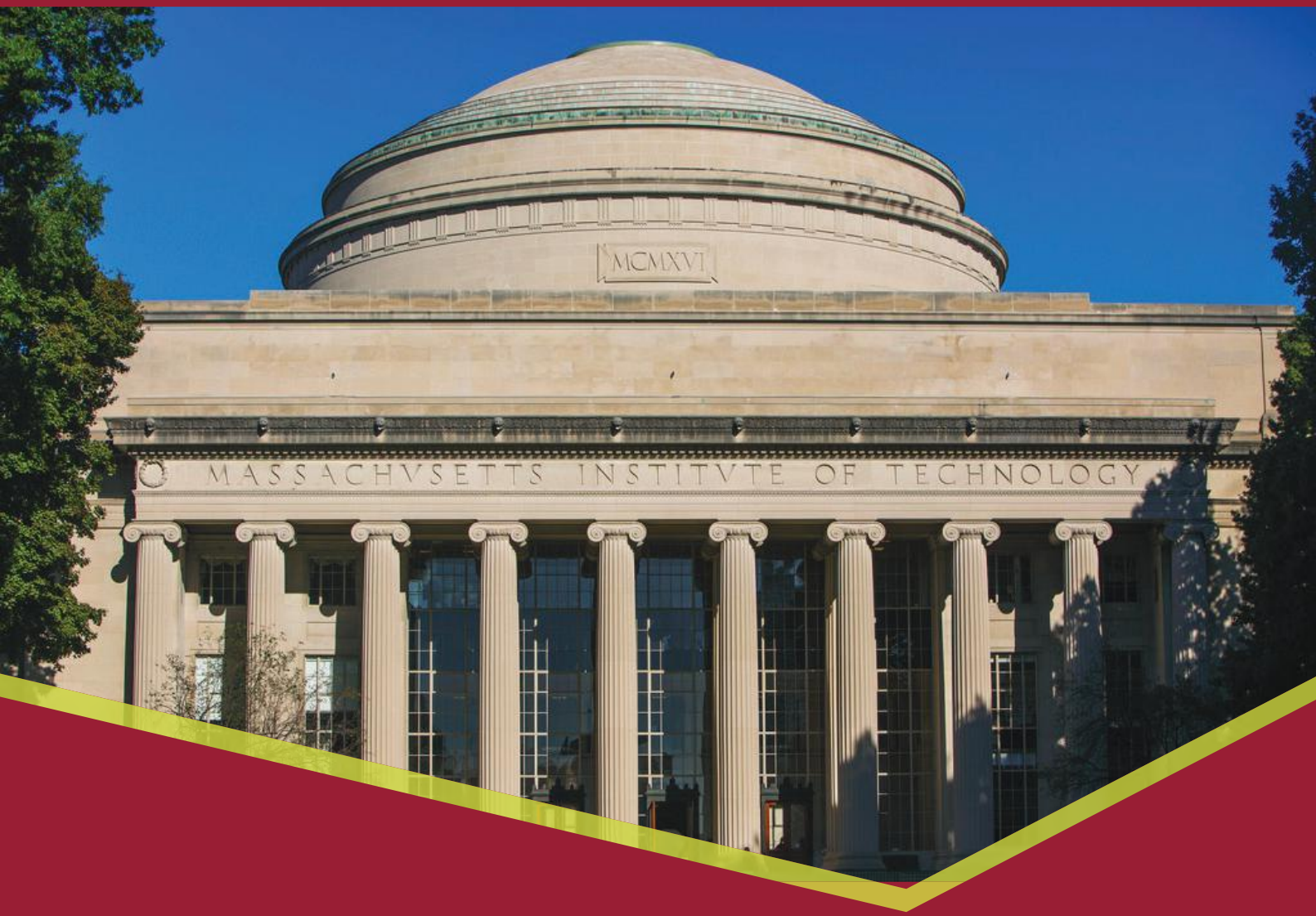
AWARDS AND HONORS

100 NOBEL LAUREATES	**6** PULITZER PRIZES	**81** MACARTHUR FELLOWSHIPS	**21** JOHN BATES CLARK MEDALS
37 FULBRIGHT SCHOLARS	**60** NATIONAL MEDAL OF SCIENCE WINNERS	**30** NATIONAL MEDAL OF TECHNOLOGY AND INNOVATION WINNERS	**16** A.M. TURING AWARD WINNERS



WE INVITE YOU TO LEARN MORE ABOUT AI AND ML: LEADING BUSINESS GROWTH

MIT Professional Education has collaborated with Northwest Executive Education to build AI and ML: Leading Business Growth into a significant learning experience for participants. Northwest works closely with renowned global universities and schools to design programs and curriculums, and to manage outreach, and delivery of these programs to participants to offer relevant, contextual, and global learning experiences.



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